# A COUNTEREXAMPLE TO SUN'S CONJECTURE ON THREE CONSECUTIVE POLYGONAL NUMBERS

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Abstract. Zhi-Wei Sun conjectured that, for every integer $m > 2$, all sufficiently large integers are sums of an $(m+1)$-gonal, an $(m+2)$-gonal, and an $(m+3)$-gonal number with nonnegative indices. We disprove this statement at $m = 33$. For every $k \geq 0$, the positive integer

$$n_k = \frac{1019^{160k+2} - 392089}{35904}$$

is not a sum of a 34-gonal, a 35-gonal, and a 36-gonal number, even when the indices are allowed to be arbitrary integers. The proof combines a factorization of an associated quadratic form with the parity of prime valuations in norms from $\mathbb{Q}(\sqrt{-2})$, and uses no theorem on primes in arithmetic progressions. A companion Lean file proves the negation of the exact statement in the Formal Conjectures repository, with no admitted proof steps.

## 1. Introduction

For an integer $j \geq 3$, define

$$p_j(t) = \frac{(j-2)t(t-1)}{2} + t \qquad (t \in \mathbb{Z}). \tag{1}$$

When $t \geq 0$, this is the usual $t$-th $j$-gonal number. In a comment on OEIS A160324 [2] dated August 21, 2009, Sun proposed that, for every integer $m > 2$, each sufficiently large integer has the form

$$p_{m+1}(x) + p_{m+2}(y) + p_{m+3}(z), \qquad x, y, z \in \mathbb{N}. \tag{2}$$

The same assertion appears in Remark 1.6 of [1] and is recorded as `OeisA160324.conjecture5` in [3]. Here and below, $0 \in \mathbb{N}$.

Our result concerns the quantifier "sufficiently large" in (2): a single small exception would not suffice. We construct an explicit unbounded sequence of exceptions for one fixed value of $m$.

**Theorem 1.1.** *For every $k \in \mathbb{N}$, the number*

$$n_k = \frac{1019^{160k+2} - 392089}{35904} \tag{3}$$

*is a positive integer. There do not exist $x, y, z \in \mathbb{Z}$ such that*

$$n_k = p_{34}(x) + p_{35}(y) + p_{36}(z).$$

*Moreover $n_k \longrightarrow \infty$. Consequently, Sun's A160324 Conjecture 5 is false.*

The obstruction comes from the binary norm $u^2 + 2v^2$. After completing squares, a hypothetical representation gives an identity $AB = 544(u^2 + 2v^2)$. One factor is eight times an integer congruent to 7 modulo 8. This forces a prime congruent to 5 or 7 modulo 8 to occur to an odd exponent in that factor. The construction prevents the prime from dividing either the other factor or 544. Quadratic reciprocity forces its exponent in the norm to be even, a contradiction.

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## 2. Two elementary prime-valuation facts

For a prime $\ell$ and a nonzero integer $w$, write $v_\ell(w)$ for the exponent of $\ell$ in $|w|$.

**Lemma 2.1.** *If $\ell$ is a prime with $\ell \equiv 5$ or $7 \pmod 8$, then $v_\ell(u^2+2v^2)$ is even for every $u, v \in \mathbb{Z}$ with $u^2+2v^2 \neq 0$.*

*Proof.* The supplementary laws of quadratic reciprocity give

$$\left(\frac{-2}{\ell}\right) = -1 \qquad (\ell \equiv 5, 7 \pmod 8).$$

If $\ell \mid u^2+2v^2$ and $\ell \nmid v$, then $(uv^{-1})^2 \equiv -2 \pmod \ell$, which is impossible. Thus $\ell \mid v$, and then $\ell \mid u$ as well.

Choose the largest $r \geq 0$ such that $\ell^r$ divides both $u$ and $v$. At least one of $u/\ell^r$ and $v/\ell^r$ is not divisible by $\ell$, so their norm is not divisible by $\ell$. Hence $v_\ell(u^2+2v^2) = 2r$. □

**Lemma 2.2.** *Every positive integer $a \equiv 7 \pmod 8$ has a prime divisor $\ell \equiv 5$ or $7 \pmod 8$ for which $v_\ell(a)$ is odd.*

*Proof.* The integer $a$ is odd. Every odd square is 1 modulo 8, so its residue modulo 8 is the product of the residues of those primes occurring to odd exponents. If these primes were all 1 or 3 modulo 8, that product would be 1 or 3 modulo 8, since $3^2 \equiv 1 \pmod 8$. This contradicts $a \equiv 7 \pmod 8$. □

## 3. The quadratic-form obstruction

The next proposition isolates the part of the proof responsible for nonrepresentation. It permits a somewhat wider choice of parameters than the sequence in Theorem 1.1.

**Proposition 3.1.** *Let $T$ be a positive integer satisfying $T \equiv 27 \pmod{32}$, and suppose that every prime divisor of $T$ is 1 or 3 modulo 8. There are no integers $X, Y, Z$ satisfying*

$$T^2 = 561X^2 + 136Y^2 + 2112Z^2, \qquad X \equiv 17 \pmod{32}, \qquad Y \equiv 1 \pmod 2. \tag{4}$$

*Proof.* Assume (4). Set

$$A = 2T + 34X + 64Z, \qquad B = 2T - 34X - 64Z.$$

A direct calculation using (4) gives

$$\begin{aligned} AB &= 4T^2 - (34X + 64Z)^2 \\ &= 544\big(Y^2 + 2(X - 2Z)^2\big). \end{aligned} \tag{5}$$

Since $Y$ is odd, the right side is positive. Also $A + B = 4T > 0$. It follows that $A, B > 0$.

The prescribed congruences imply

$$A \equiv 2 \cdot 27 + 34 \cdot 17 \equiv 56 \pmod{64}.$$

Thus $A = 8a$ for a positive integer $a \equiv 7 \pmod 8$. By Lemma 2.2, there is a prime $\ell \equiv 5$ or $7 \pmod 8$ with $v_\ell(a)$ odd. This prime is odd, so $v_\ell(A)$ is odd as well.

If $\ell$ divided $B$, then it would divide $A + B = 4T$. As $\ell$ is odd, it would divide $T$, contrary to the hypothesis on the prime divisors of $T$. Hence $\ell \nmid B$. Furthermore $\ell \nmid 544 = 2^5 \cdot 17$, since $17 \equiv 1 \pmod 8$. Taking $\ell$-adic valuations in (5) therefore yields

$$v_\ell(A) = v_\ell\big(Y^2 + 2(X - 2Z)^2\big).$$

The left side is odd; the right side is even by Lemma 2.1. This is a contradiction. □

## 4. An unbounded family of exceptions

*Proof of Theorem 1.1.* Put $D = 35904$ and $C = 392089$. The following identities and congruences will be used:

$$D = 64 \cdot 3 \cdot 11 \cdot 17, \qquad 1019^2 - C = 18D, \qquad 1019^{80} \equiv 1 \pmod{D}. \tag{6}$$

For completeness, the last congruence can be checked separately modulo the four pairwise coprime factors of $D$. Modulo 64, we have $1019 \equiv -5$ and $(-5)^{16} \equiv 1$. Modulo 3, 11, and 17, the respective residues are $-1$, 7, and $-1$, with orders dividing 2, 10, and 2. These orders all divide 80, so the Chinese remainder theorem proves the assertion.

For $k \geq 0$, set $T = 1019^{80k+1}$. By (6),

$$T \equiv 1019 \pmod{D}, \qquad T^2 \equiv C \pmod{D}. \tag{7}$$

Consequently $n_k = (T^2 - C)/D$ is an integer. It is positive because $T \geq 1019$ and $1019^2 - C = 18D > 0$. In particular $n_0 = 18$. Since $T \to \infty$, we also have $n_k \to \infty$.

The integer 1019 is prime: the primes below $32 > \sqrt{1019}$ are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31$, and none divides it. Thus every prime divisor of $T$ equals $1019 \equiv 3 \pmod 8$. Since $32 \mid D$, (7) also gives $T \equiv 1019 \equiv 27 \pmod{32}$.

Suppose that $n_k = p_{34}(x) + p_{35}(y) + p_{36}(z)$ for integers $x, y, z$. Define

$$X = 32x - 15, \qquad Y = 66y - 31, \qquad Z = 17z - 8.$$

Expanding (1), or completing squares, gives

$$D\big(p_{34}(x) + p_{35}(y) + p_{36}(z)\big) + C = 561X^2 + 136Y^2 + 2112Z^2. \tag{8}$$

Its left side is $Dn_k + C = T^2$. Moreover $X \equiv 17 \pmod{32}$ and $Y$ is odd. Proposition 3.1 now gives a contradiction. This rules out representations with arbitrary integer indices and therefore also those with nonnegative indices.

The sequence $n_k$ is unbounded, so no bound can make (2) hold for every larger integer at $m = 33$. □

*Remark* 4.1. The same argument applies to any prime $T \equiv 3 \pmod 8$ with $T^2 \equiv C \pmod D$ and $T \equiv 27 \pmod{32}$, provided $T^2 > C$. Some smaller examples are displayed in Table 1. They illustrate the obstruction; the explicit prime-power family is what proves infinitely many exceptions without an appeal to Dirichlet's theorem.

Table 1. Additional exceptions, with $n = (T^2 - 392089)/35904$.

| $T$ | $n$ | $T$ | $n$ |
|---|---|---|---|
| 1019 | 18 | 23291 | 15098 |
| 4283 | 500 | 28219 | 22168 |
| 8059 | 1798 | 36923 | 37960 |
| 10267 | 2925 | 41243 | 47365 |

## 5. Formal statement and scope

The companion file is named `SunConjecture5.lean`. It imports the original problem module and uses the repository's `polygonalNumber` definition. In that definition, the expression $x(x-1)/2$ is natural-number division. The connection with (1) is exact for natural $x$: the product $x(x-1)$ is even, and the case $x = 0$ is included explicitly.

The final formal theorem, `sun_conjecture5_false`, proves the negation of the complete type of `OeisA160324.conjecture5`, including its universal quantifier over $m > 2$ and its eventual quantifier over $n$. The expression `type_of% OeisA160324.conjecture5` extracts that type without invoking the repository's admitted proof. The proof exhibits arbitrarily large counterexamples at $m = 33$.

The single Lean file also proves the closed formula (3), the positivity of every $n_k$, the bound $n_k \geq k$, and nonrepresentation for arbitrary integer indices. It was checked with Lean 4.33.1 and the repository's pinned Mathlib revision `0df444a360eaa60ab8c11dca51a86af692955474`, against the original problem module at the commit recorded in [3]. The kernel check succeeds. The command

```
#print axioms SunConjecture5.sun_conjecture5_false
```

reports precisely `propext`, `Classical.choice`, and `Quot.sound`, the standard axioms used by this Mathlib proof. Neither the final theorem nor the theorem for integer indices depends on `sorryAx`, a custom axiom, or a native-evaluation axiom.

This result concerns only the eventual three-term assertion in (2). It does not settle the square–pentagonal–hexagonal universality statement also listed under A160324, or the assertion that allows an extra remainder between 0 and $m - 3$.

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## References

[1] Z.-W. Sun, *On universal sums of polygonal numbers*, Science China Mathematics **58** (2015), 1367–1396. arXiv:0905.0635; see Remark 1.6.

[2] The OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, sequence A160324, comment by Z.-W. Sun dated August 21, 2009. https://oeis.org/A160324.

[3] Google DeepMind and contributors, *Formal Conjectures*, `FormalConjectures/OEIS/160324.lean`, `OeisA160324.conjecture5`, commit `89294ea02bd7cd678d59984add52cb4baef3dbf4`. https://github.com/google-deepmind/formal-conjectures.

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